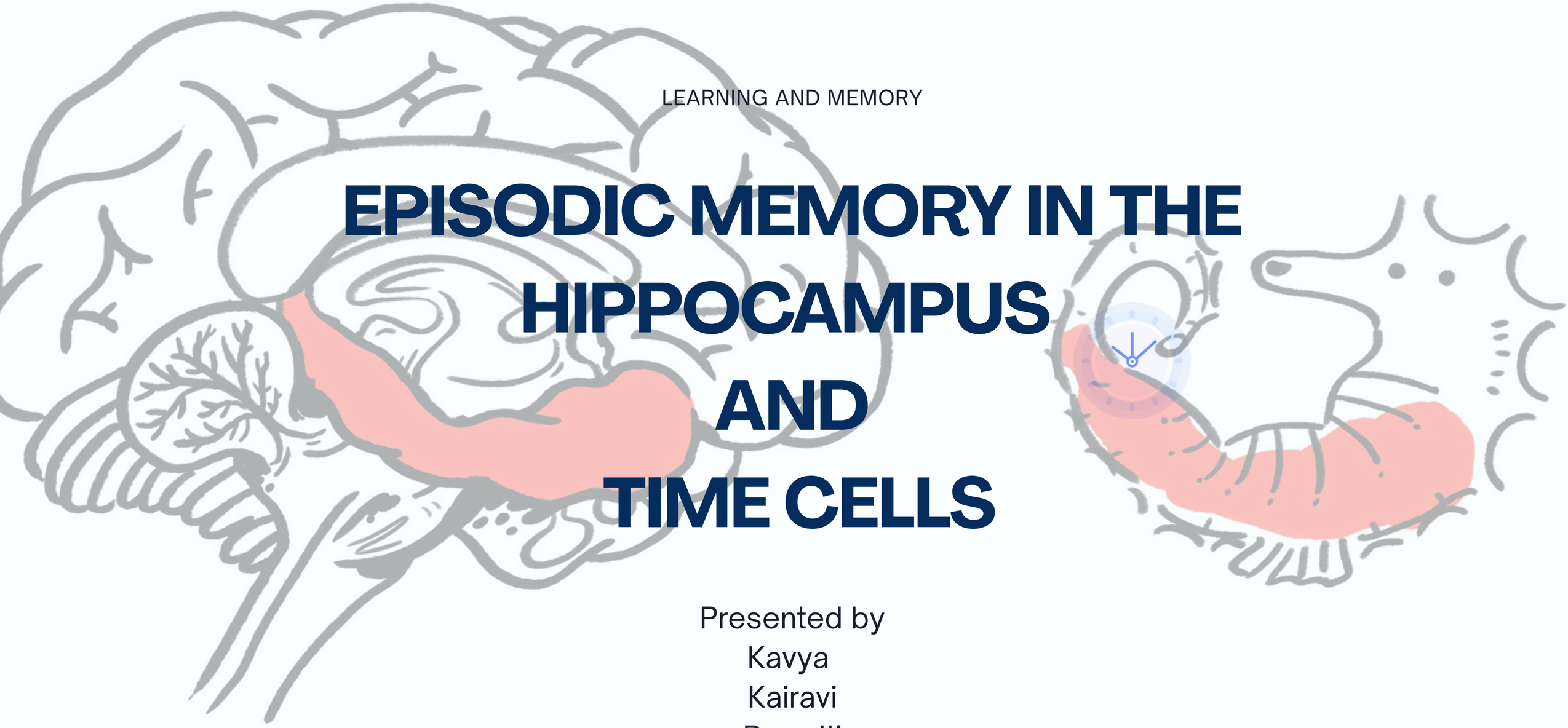
LEARNING AND MEMORY

EPISODIC MEMORY IN THE HIPPOCAMPUS AND TIME CELLS



Presented by
Kavya
Kairavi
Pravalli

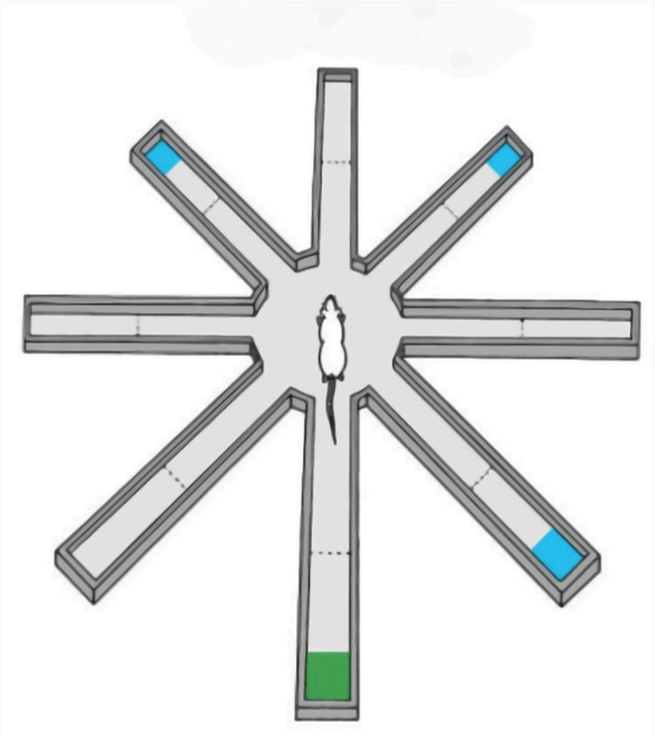
The Parking Lot Problem



A hungry rat must visit eight arms to find food. Once an arm is cleared, re-entering it is an error.

The "When" and "Where"
Success requires "episodic-like" memory the ability to bind the what (food) to a specific where (arm number) and a specific when (visited during this trial, not a previous one).

Healthy rats quickly learn to clear the maze with zero re-entries. However, rats with hippocampal lesions fail miserably. Lesioned rats aren't "forgetful"; they know food is in the arms, but they cannot distinguish between an arm visited five minutes ago and an arm visited yesterday.



Radial Arm Maze



Episodic memory is oriented to the past in a way in which no other kind of memory, or memory system,is. It is the only memory system that allows people to consciously re-experience past experiences. Its special, and unique, relationship to time, surprisingly, is not widely known. Nor is it, I think, adequately appreciated. Most people naturally associate all memory with the past and are astonished to learn that this is not so.

-Eden Tulving 2002

EPISODIC MEMORY: From Mind to Brain, 2002

The case of K.C.

Intact Semantic Memory, Destroyed Episodic Memory

Complete loss of autonoetic (re-experience) consciousness.
Mentally trapped in the immediate present.

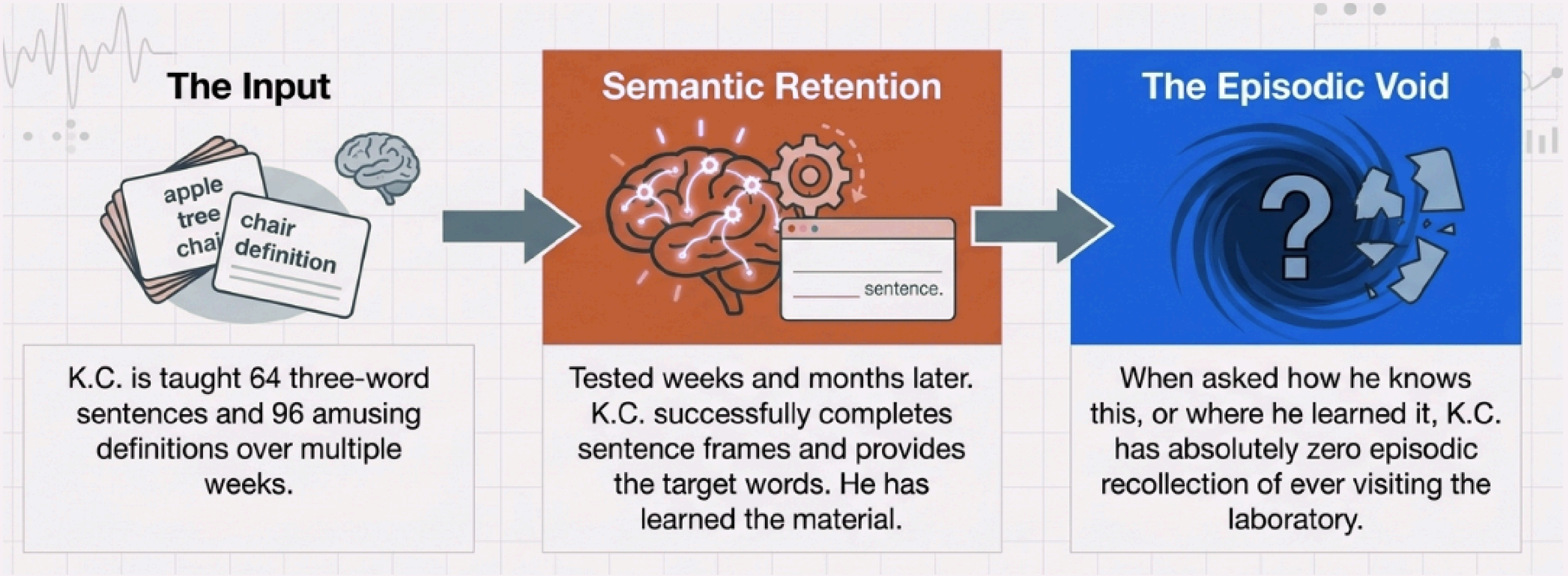
- Plays chess, cannot recall a single personal event from birth to present moment.
- Knows where exactly his family's summer cottage is but denies of having ever been there



Figure 1 K.C., a man who lost his episodic memory as a result of traumatic brain injury in 1981, photographed in 1986 when he was 35 years of age.

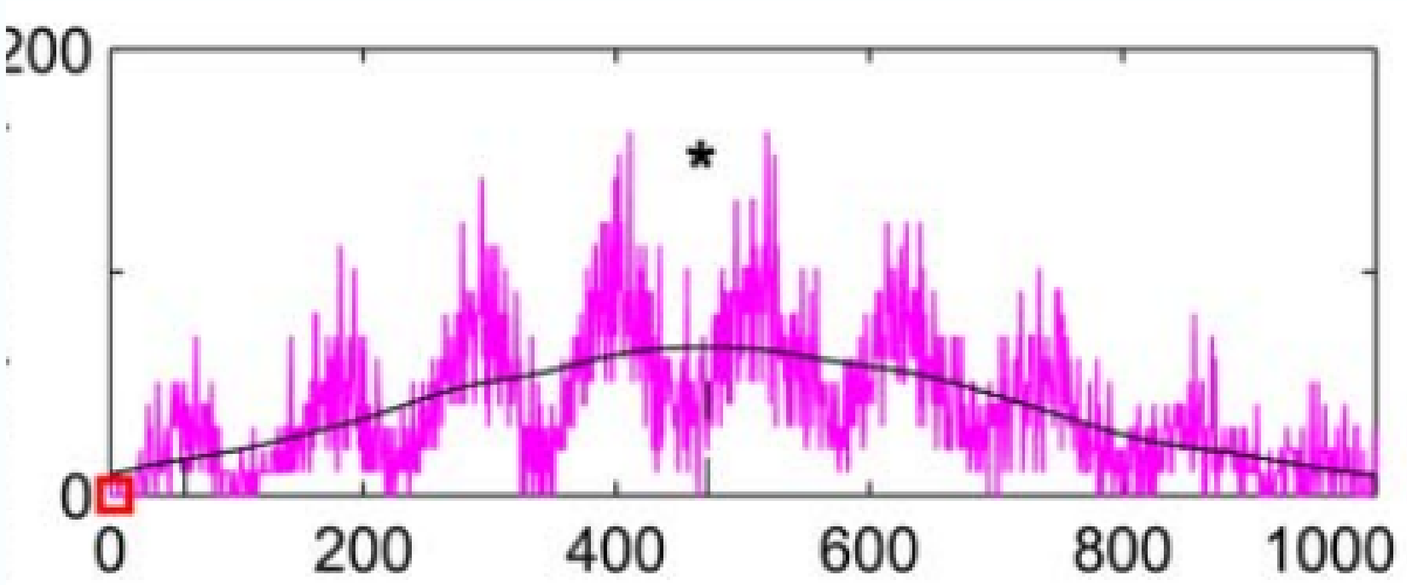
- Has zero memory of being evacuated from his home along with tens of thousands of people during a deadly chemical train derailment.

Independence of Memory Systems



Subjective time is a bidirectional continuum

How do you know ‘when’ the ‘what’ happened?



Theta Oscilations

Dragoi and Buzsáki (2006) showed that the hippocampus uses theta sequences to compress experience.

Dragoi and Buzsáki (2006)

Focus on Theta Sequences—the high-speed "temporal compression" of spatial paths.

As a rat runs through a sequence of locations (A → B → C), the corresponding place cells fire in that exact order.

However, they don't just fire as the rat physically enters the location. Instead, within a single cycle of a theta wave (~125 milliseconds), the hippocampus fires a compressed version of the sequence, representing **where the rat was, where it is now, and where it is going.**

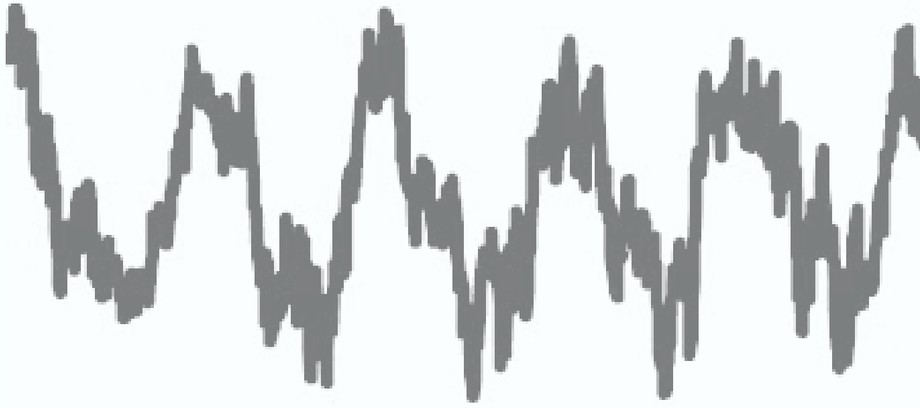
This brings the firing of neurons close enough together in time (milliseconds) to trigger Synaptic Plasticity. Without this compression, the brain couldn't "wire" the sequence together because the real-world events are too far apart.

The phase of the theta oscillation is what organizes these spikes

This provides a "temporal framework" that ensures the order of events is preserved, preventing the "What" from getting disconnected from the "When".



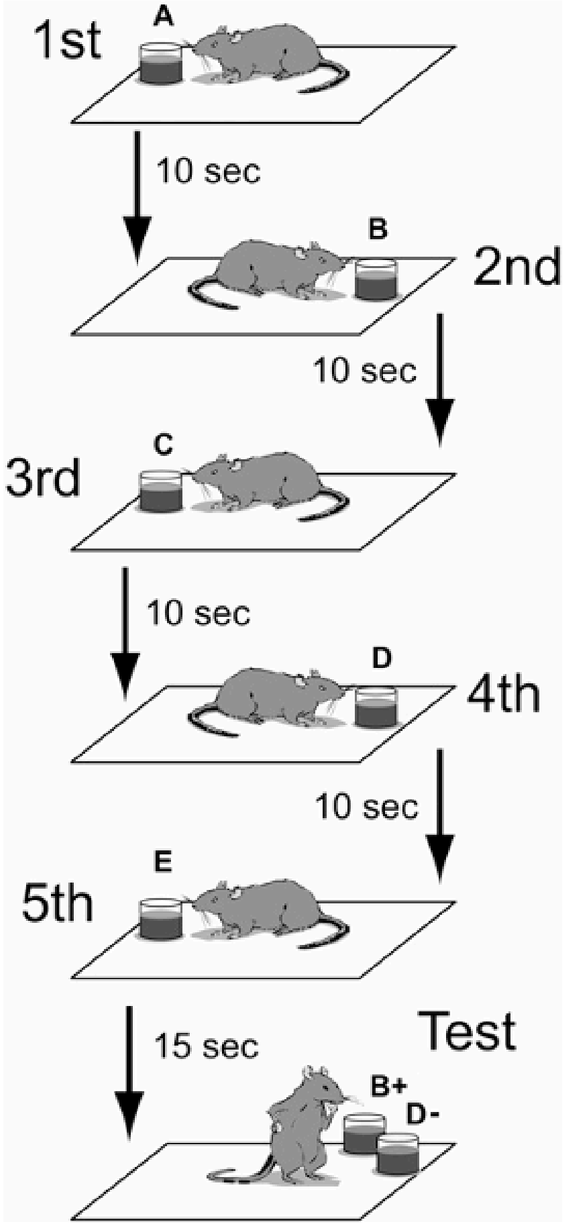




Hippocampal Representation of Temporal Order
Manns et al., 2007

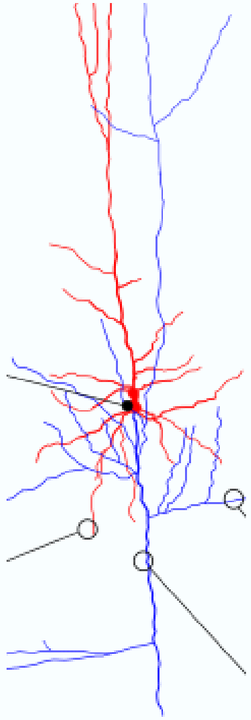
The Odor Sequence Task: Much like the radial maze, this task requires rats to remember the order of odors (A → B → C → D → E).





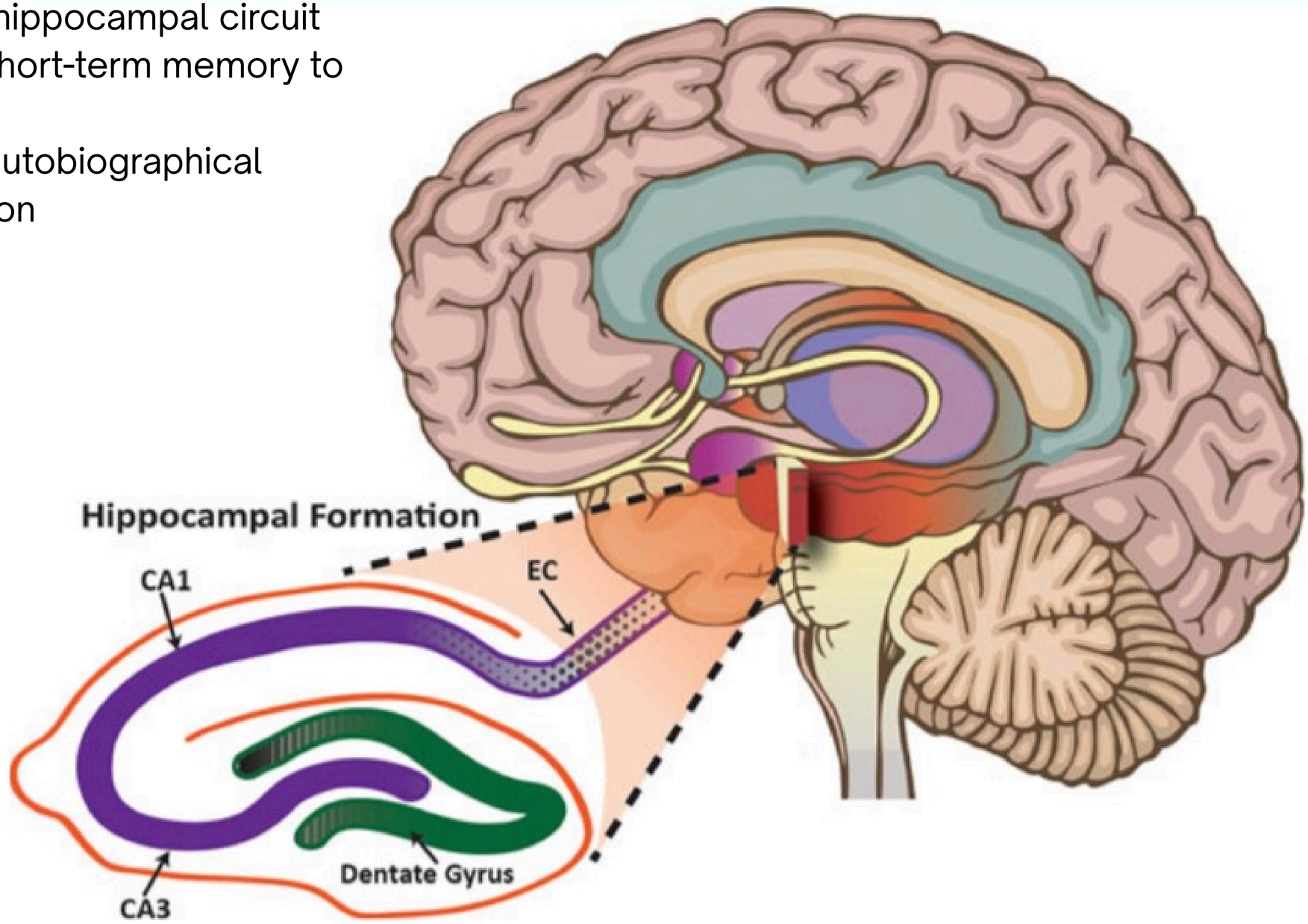
CA1

- Primary output hub of the hippocampal circuit
- Critical for consolidating short-term memory to long-term storage
- Retrieval of episodic and autobiographical memories, novelty detection
- Spatial coding



Pyramidal neurons

Medial Temporal Lobe

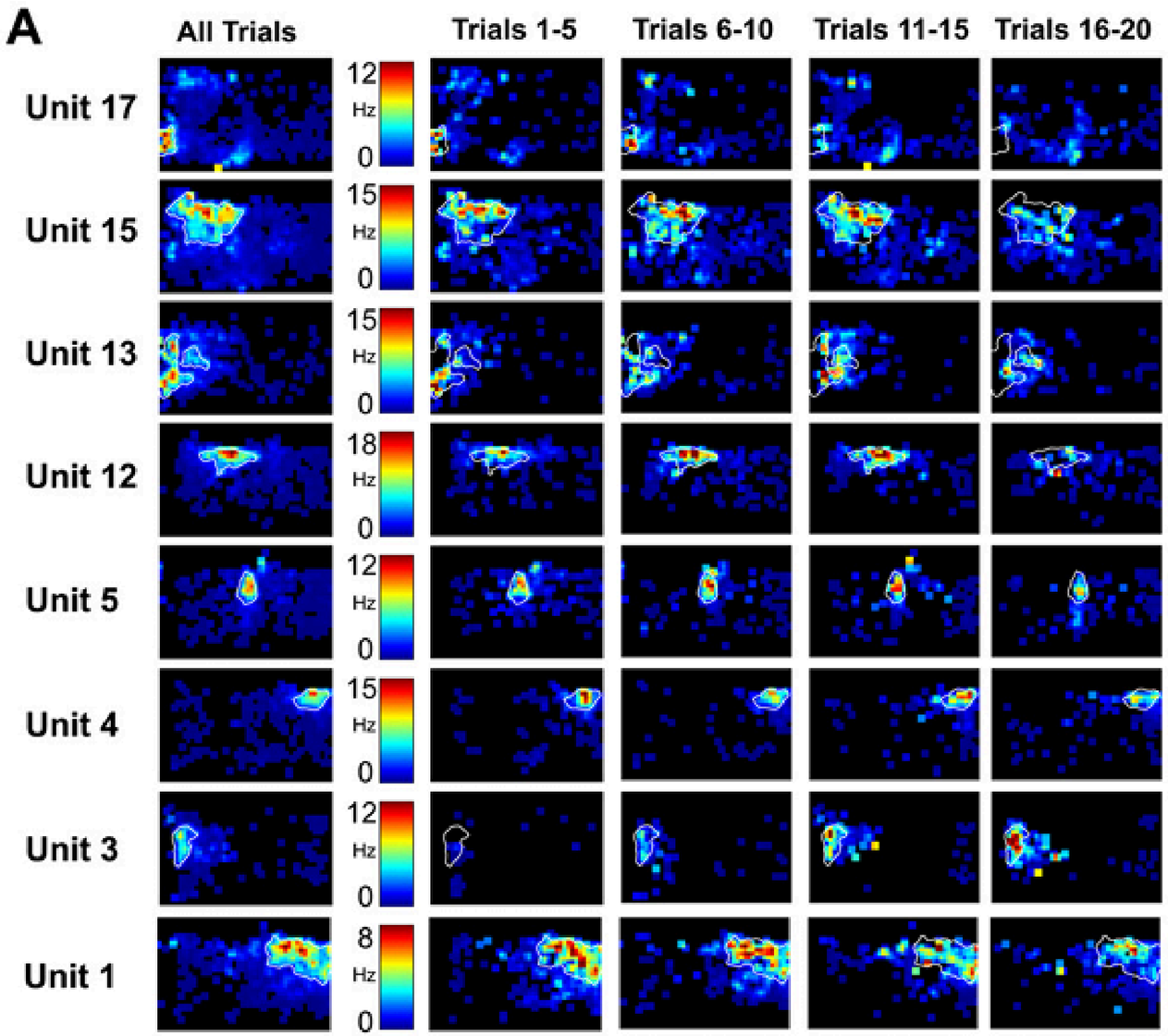


Hippocampal Representation of Temporal Order
Manns et al., 2007

Gradual Neural Drift
Even when the rat is in the exact same location, the hippocampal representation is changing over time
This drift provides the "temporal context" needed to distinguish between events that are spatially identical but occur at different times.

Using the Distance Index, they proved that even if a rat is in the same physical location, the neural pattern "drifts" over time.

Hippocampal population code is never static. It is in a constant state of gradual change.

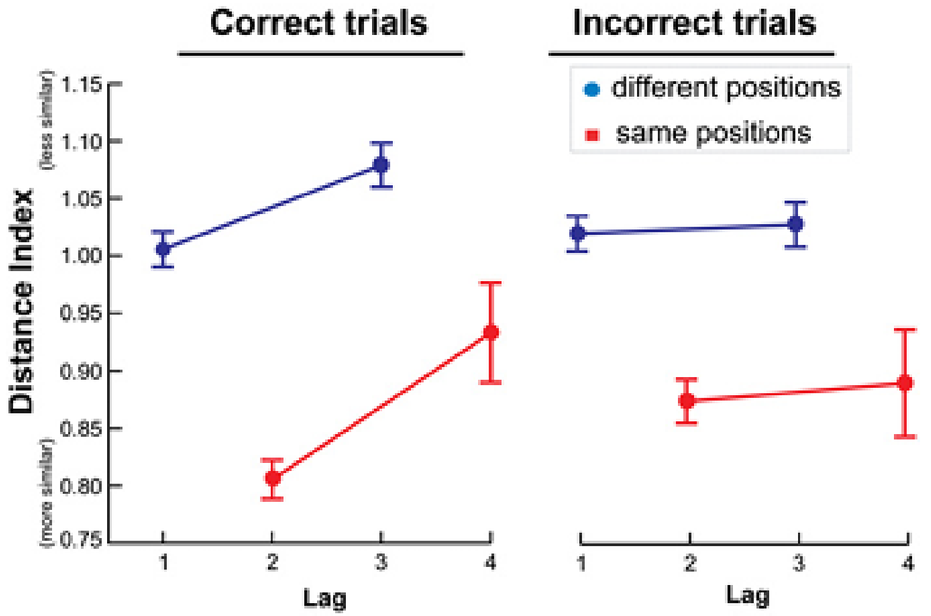


Hippocampal Representation of Temporal Order
Manns et al., 2007

When measuring the Distance Index across the entire session, events that occurred in the **same physical location triggered vastly more similar neural patterns** than events in different locations

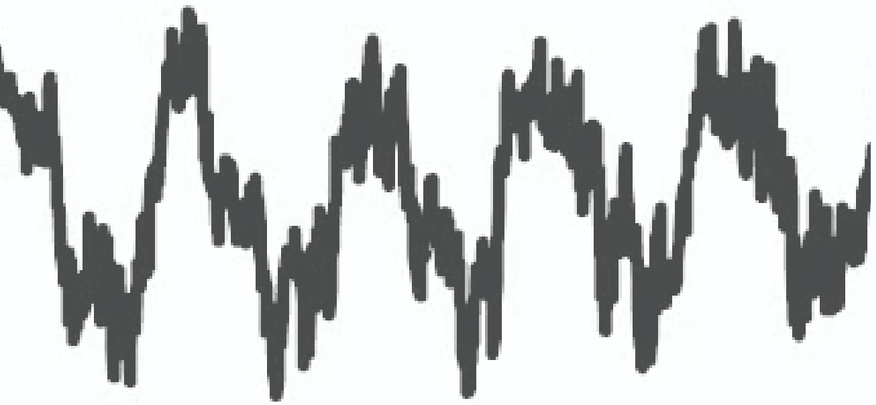
Events that happened **closer together in time and space shared similar neural signatures**

What happens when the rat forgets the sequence of odors?



On incorrect trials, the spatial map remains perfectly intact but the temporal gradient completely vanishes.
Memory for sequence order requires this gradually changing temporal context.

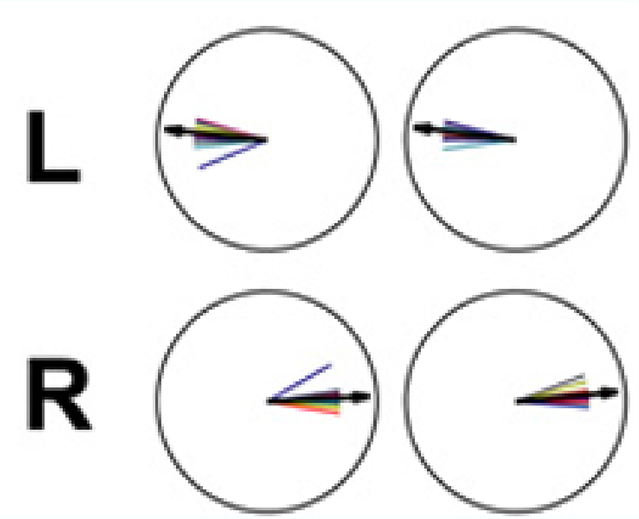
What explains the rats' ability to remember the odor sequence?



Mean Theta Duration?



Mean Running Speed?



Mean Running Direction?

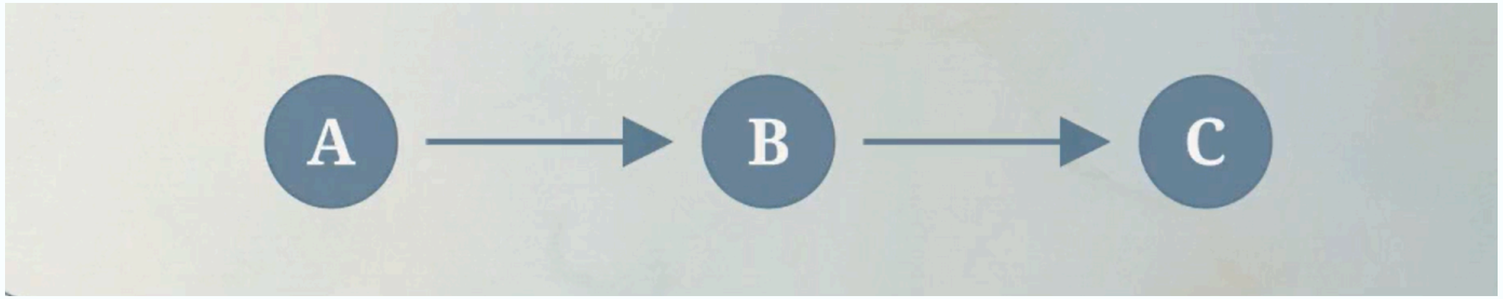
thorough analysis found no evidence that activity in the hippocampus contributed to performance in either of these possible ways

Hippocampal Representation of Temporal Order
Manns et al., 2007

- The hippocampus reflects temporal context through a gradual, continuous drift in the pattern of activity

- The temporal context is not generated by a single internal timer, but is instead built from various "temporal cues" that fluctuate over time.

- Successful memory for the order of events is dependent on the hippocampus binding individual stimuli to a gradually changing background of ensemble activity that functions as a unique temporal context



References

Dragoi, G., & Buzsáki, G. (2006). Temporal encoding of place sequences by hippocampal cell assemblies. Neuron, 50(1), 145–157. https://doi.org/10.1016/j.neuron.2006.02.023

Gluck, M. A., Mercado, E., & Myers, C. E. (2016). Learning and memory: From brain to behavior (3rd ed.). Worth Publishers.

Manns, J. R., Howard, M. W., & Eichenbaum, H. (2007). Gradual changes in hippocampal activity support remembering the order of events. Neuron, 56(3), 530–540. https://doi.org/10.1016/j.neuron.2007.08.017

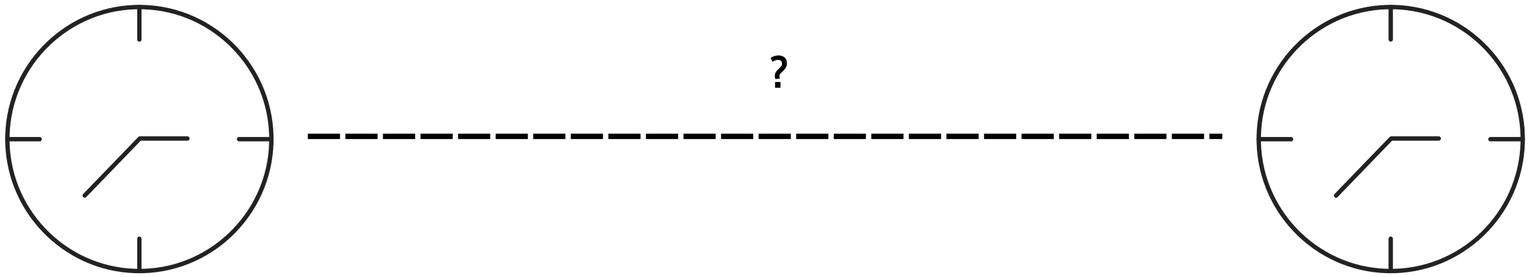
Tulving E. (2002). Episodic memory: from mind to brain. Annual review of psychology, 53, 1–25. https://doi.org/10.1146/annurev.psych.53.100901.135114

So far we've seen:
-The role of CA1 pyramidal neurons
-the hippocampus represents the order of events

Time Cells

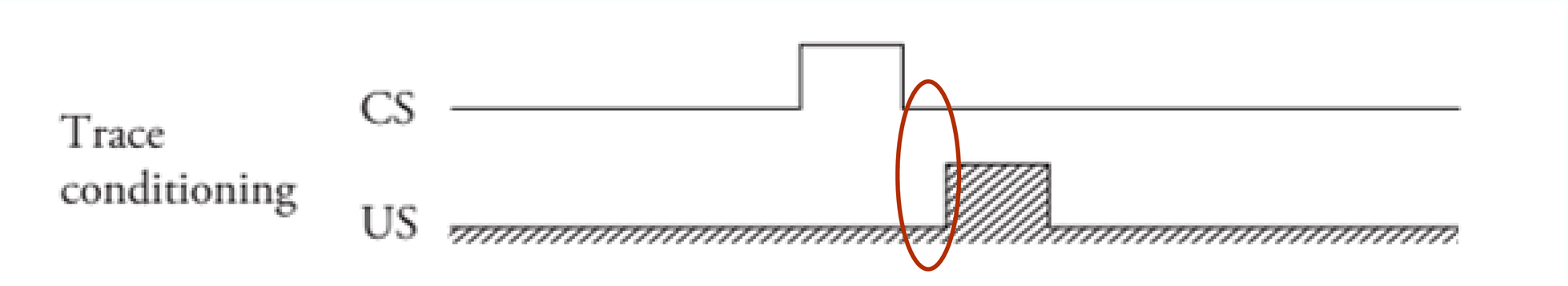
How are two discontiguous events temporally linked?

How is time represented in the hippocampus?



Background

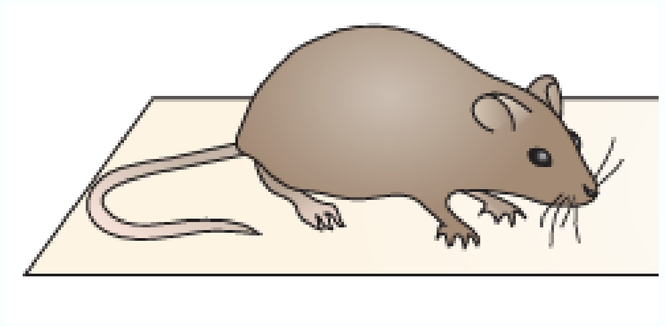
Classical Trace Conditioning



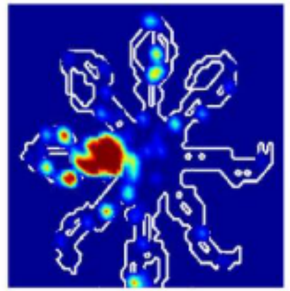
→ The US is delivered AFTER a delay after the delivery of CS. The animal STILL learns to associate the two, which means there must be a trace/memory/representation of the CS.

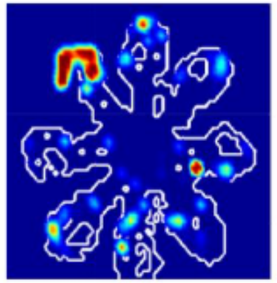
→ So how are these two apparently separate events “bridged” in our brains in order to make complex associations?

The mechanism for spatial representation?

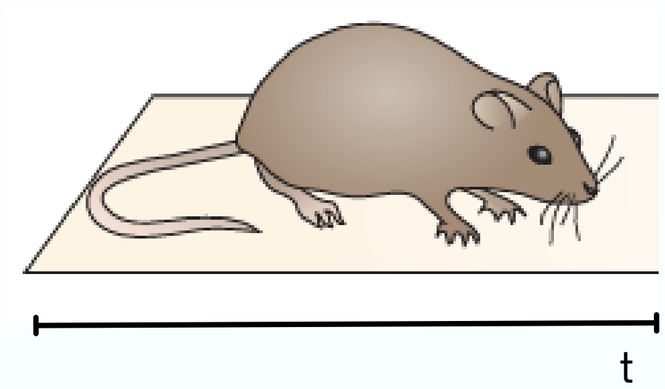


→ “place cells” are neurons in the hippocampus that fire when an animal occupies a particular location in the environment





What is the mechanism for temporal representation?



→ parallelly, evidence for “time cells” in the hippocampus: neurons that fire when an animal is in a particular MOMENT in a temporally structured experience, encoding scalar time.

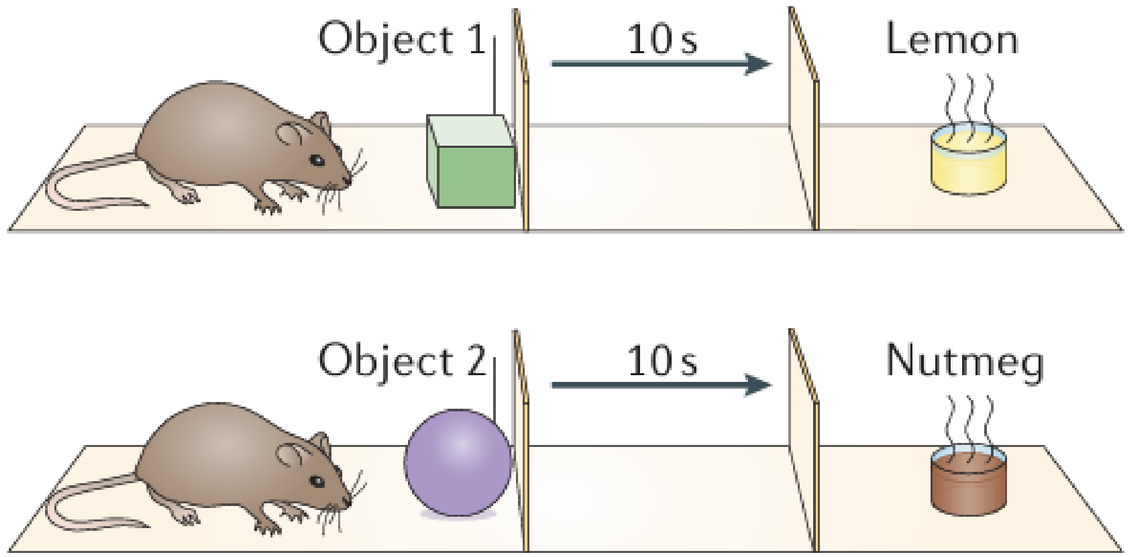
Can I convince you?



Neuron
Article

Hippocampal "Time Cells" Bridge the Gap in Memory for Discontiguous Events

Christopher J. MacDonald,[1] Kyle Q. Lepage,[1] Uri T. Eden,[1] and Howard Eichenbaum[1,*]
[1]Center for Memory and Brain, Boston University, Boston, MA 02215, USA
*Correspondence: hbe@bu.edu
DOI 10.1016/j.neuron.2011.07.012



Rats performed this **non-spatial task**
→ while tetrodes were inserted in the hippocampus, recording from isolated **pyramidal cells of dorsal CA1**

→ Rats learned to associate the 2 visually distinct objects with one of the odours.

If paired → GO = REWARD
If unpaired → NO GO = REWARD

→ Crucial question: how do hippocampal neurons represent and bridge these temporal gaps between 2 discontiguous events?

Findings from the neurons

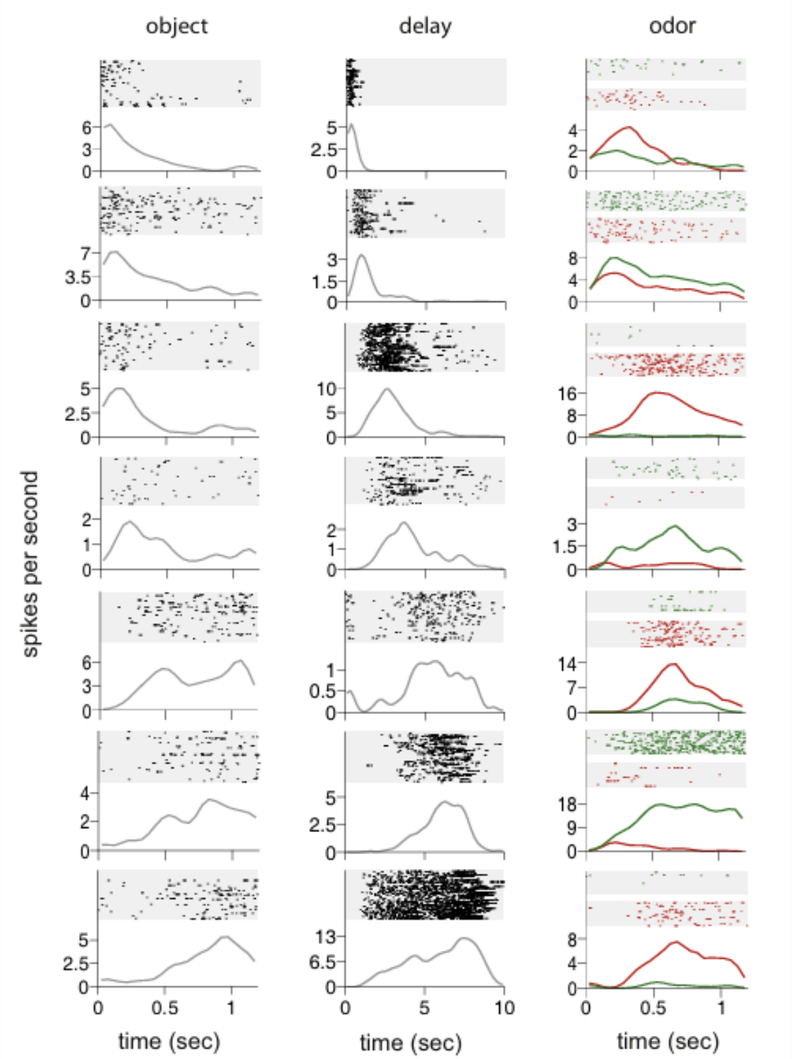


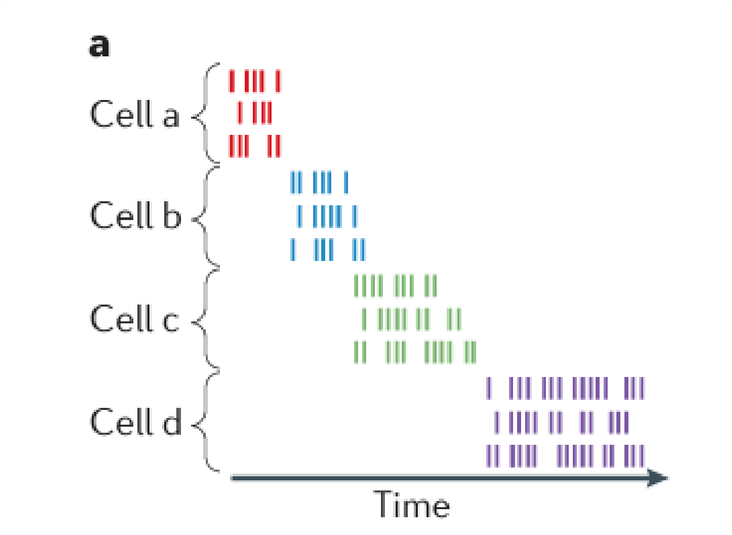
Figure 2. Hippocampal Neurons Activate at Successive Times during Each Trial Period

Out of the total neurons that activated during a trial,

1. 30% activated during the object period- their magnitude and activity of firing differed considerably

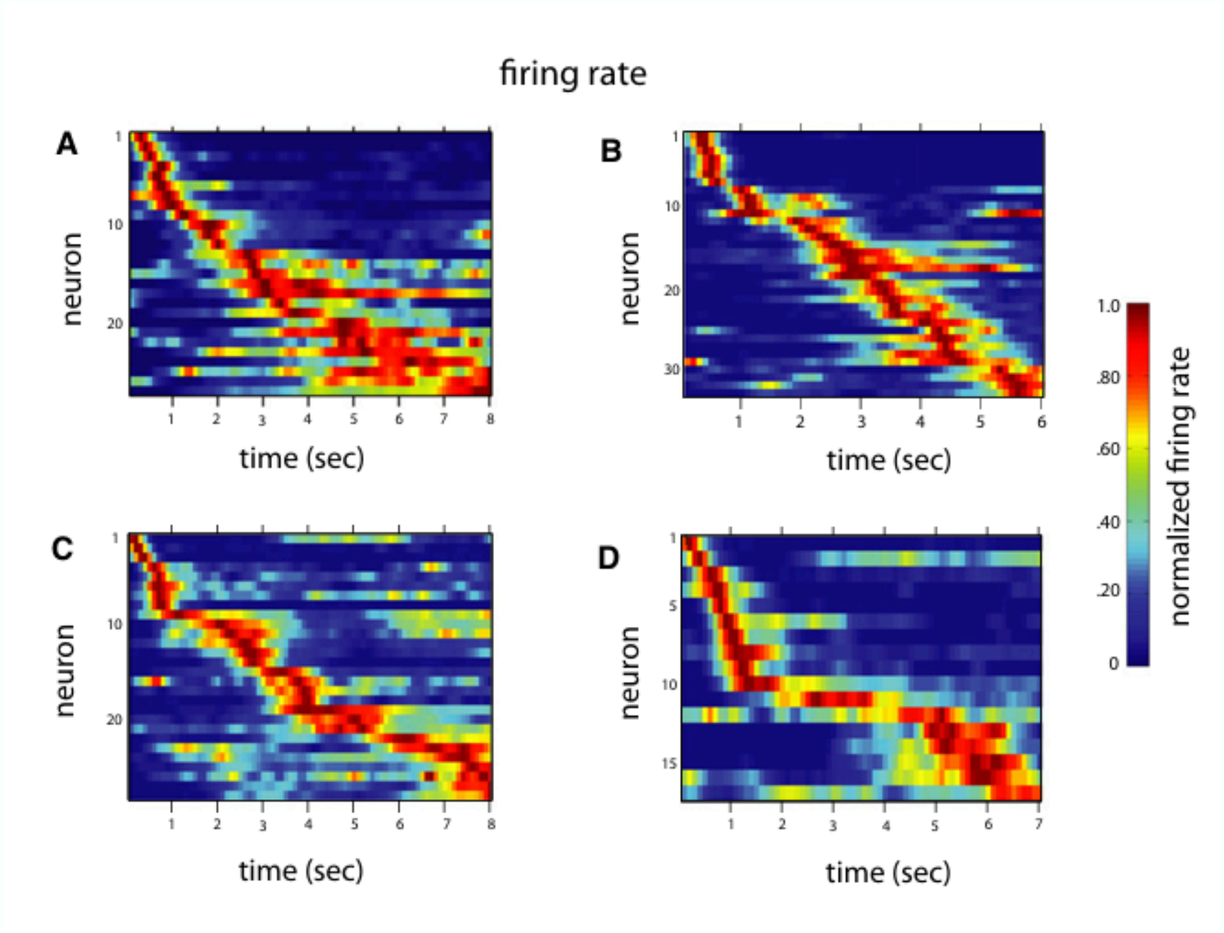
2. 28% activated during the odour period- their magnitude and activity of firing differed considerably

3. 52% activated during the delay period- showed striking selectivity to specific moments in the delay



Larger Ensembles of Neurons

averaged normalised firing rates across all trials (4 sessions) for each rat



→ The mean peak firing rate for each time cell occurred at sequential moments

→ The overlap among firing periods bridges the entire delay

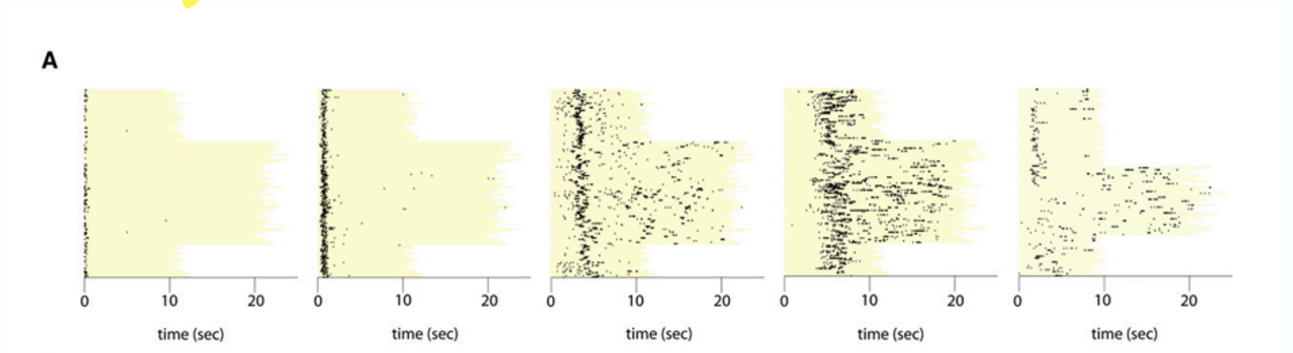
Conclusion: The firing patterns of these cells reflect the ‘passage of time’.

1. Extensive general linear model analysis: activity patterns associated with the animal's location, speed and head direction during the delay period were distinguished from activity patterns associated with the time elapsed.

2. Recent studies with head-fixed animals (including primates and humans) revealed similar results.

Strong evidence for time cells

3. the firing patterns of many of these neurons changed-- or 'retimed' when the delay was increased. This happened even though the behaviour and locations of the animal during the initial period did not change

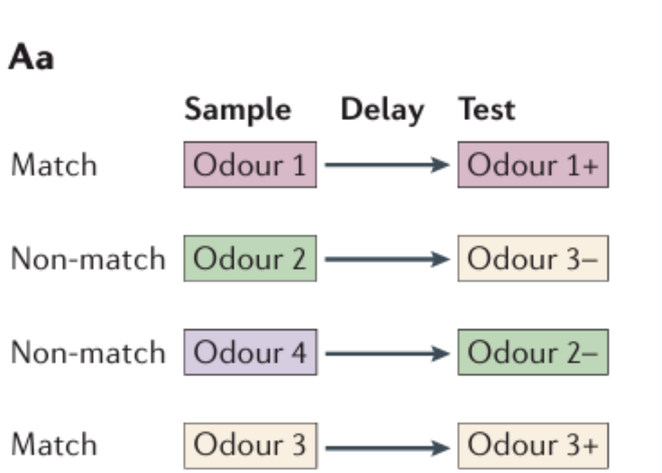


What is the role of Time Cells in Memory?

Behavioral/Cognitive

Distinct Hippocampal Time Cell Sequences Represent Odor Memories in Immobilized Rats

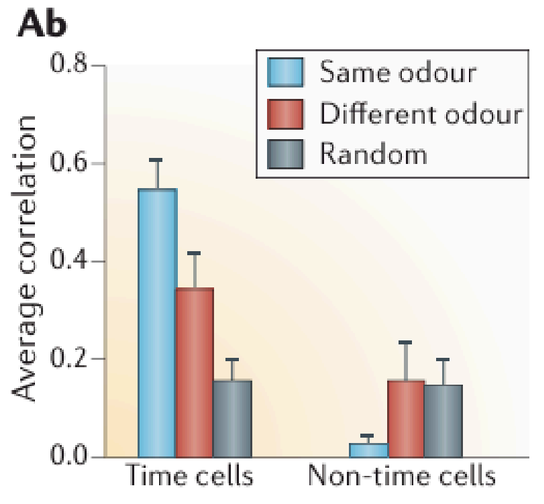
Christopher J. MacDonald, Stephen Carrow, Ryan Place, and Howard Eichenbaum
Department of Psychology and the Center for Memory and Brain, Boston University, Boston, Massachusetts 02215



Within these time cells, there is even more specificity:

Population vectors during the **delay period** for time cells:

- highest correlation between trials that began with the same sample odour
- less strong (but still above control) correlations between trials with different sample odour



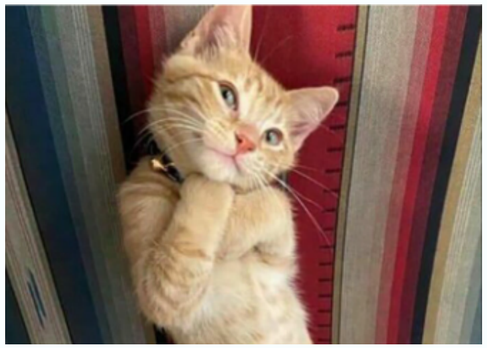
Which means that the time cell firing (during the delay) for each odour sample had a unique memory.

That is how rats ‘remember’ how to correctly respond to the test odours.

Summary and References

Larger claim: time cells encode the temporality of episodic memory, and ongoing movement through space is not necessary for temporal representations in the hippocampus (MacDonald et al., 2013)

Have I convinced you?
If not, ask away!



Howard, M. W., & Eichenbaum, H. (2014). Time cells in the hippocampus: A new dimension for mapping memories. Nature Reviews Neuroscience, 15(11), 732-744.

MacDonald, C. J., Carrow, S., Place, R., & Eichenbaum, H. (2013). Distinct hippocampal time cell sequences represent odor memories in immobilized rats. Journal of Neuroscience, 33(36), 14607-14616.

MacDonald, C. J., Lepage, K. Q., Eden, U. T., & Eichenbaum, H. (2011). Hippocampal "time cells" bridge the gap in memory for discontiguous events. Neuron, 71(4), 737-749.

MacDonald et al. showed:

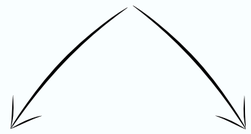
hippocampal ‘time cells’ fire sequentially across delay periods

Time Cells Are Not Enough

From milliseconds to years:

Understanding how the hippocampus encodes time across vastly different scales reveals why time cells alone cannot explain long-term memory.

small delay timeline:
early → middle → late



Seconds → Minutes → Days → Weeks → Years

How does the hippocampus encode long-term temporal context?

The Long-Term Memory Problem

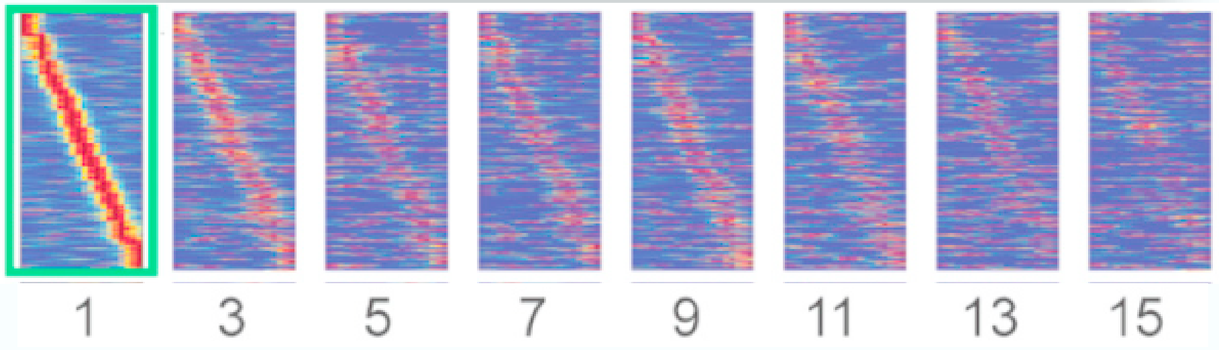
The hippocampus preserves stable spatial context while gradually evolving temporal context

Imagine attending the SAME classroom every day

DAY 1
ABCD

DAY 2
ABCE

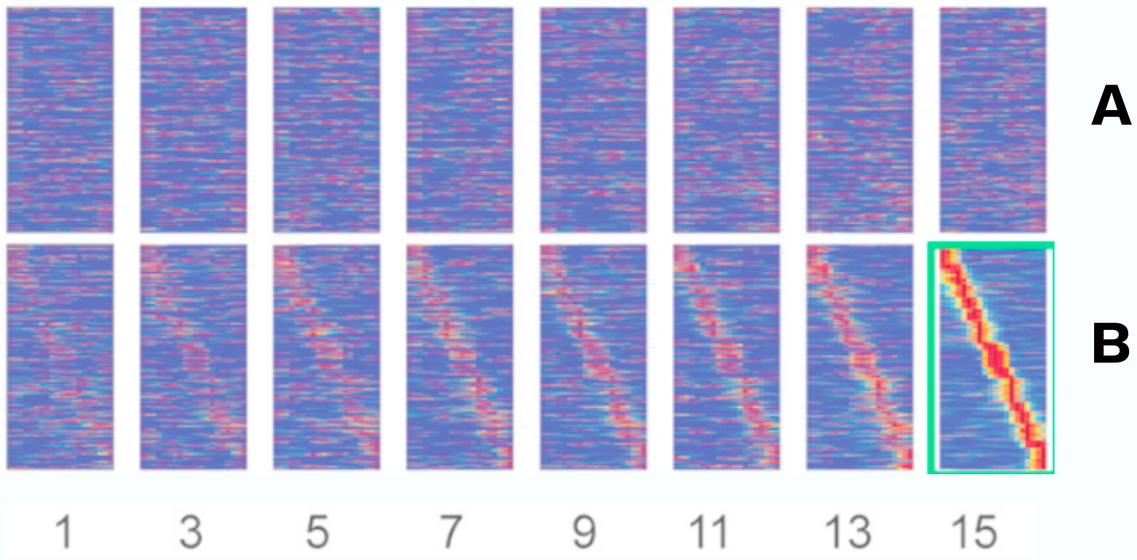
DAY 21
BEFG

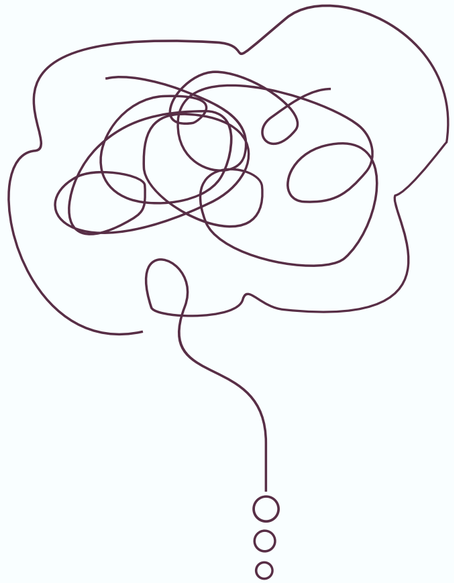


Stable spatial maps across days despite evolving ensemble activity (Rubin et al 2015)



Temporal drift within stable spatial context



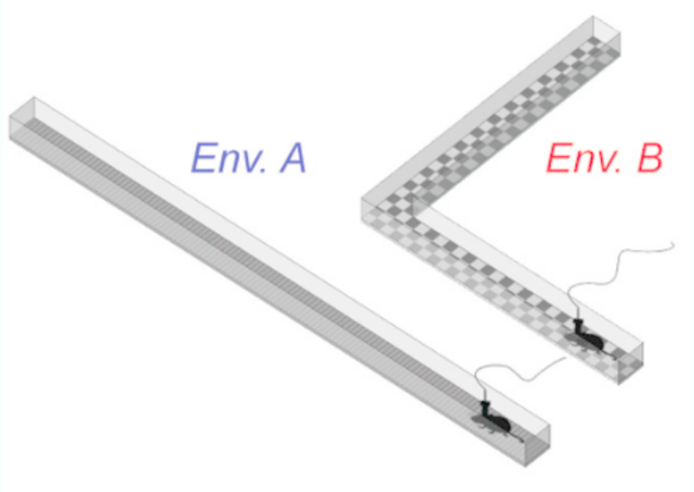


Different environments recruit distinct ensembles, while ensemble activity within each environment gradually drifts across time

Rubin, A. et al. (2015)

Now you are going to visit the SAME café everyday for coffee

DAY 1	DAY 2	DAY 21
WXYZ	**WXYQ**	**XYQR**



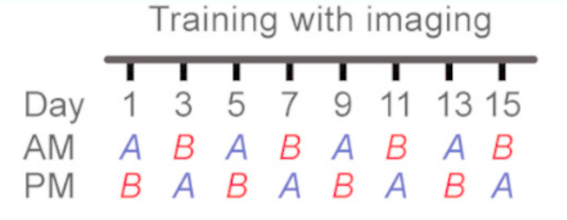
5 Mice
CA1 RECORDINGS
Longitudinal calcium imaging

Using miniature microscopes and GCaMP calcium indicators, the same hippocampal neurons were tracked across multiple days.

Two highly familiar but distinct environments

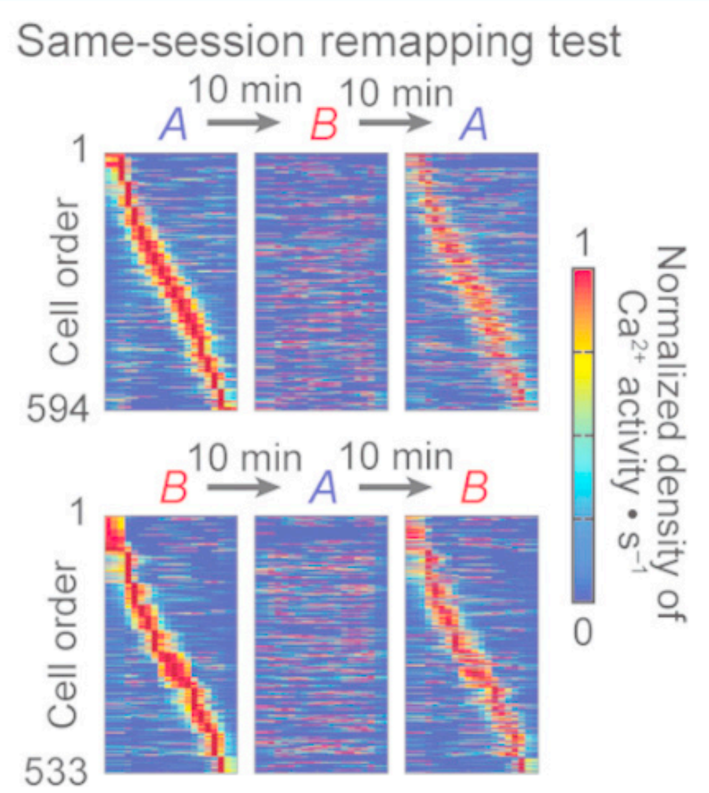
- Different spatial layouts
- Different sensory cues
- Used to distinguish spatial remapping from temporal drift

15 days
Sessions every 2 days

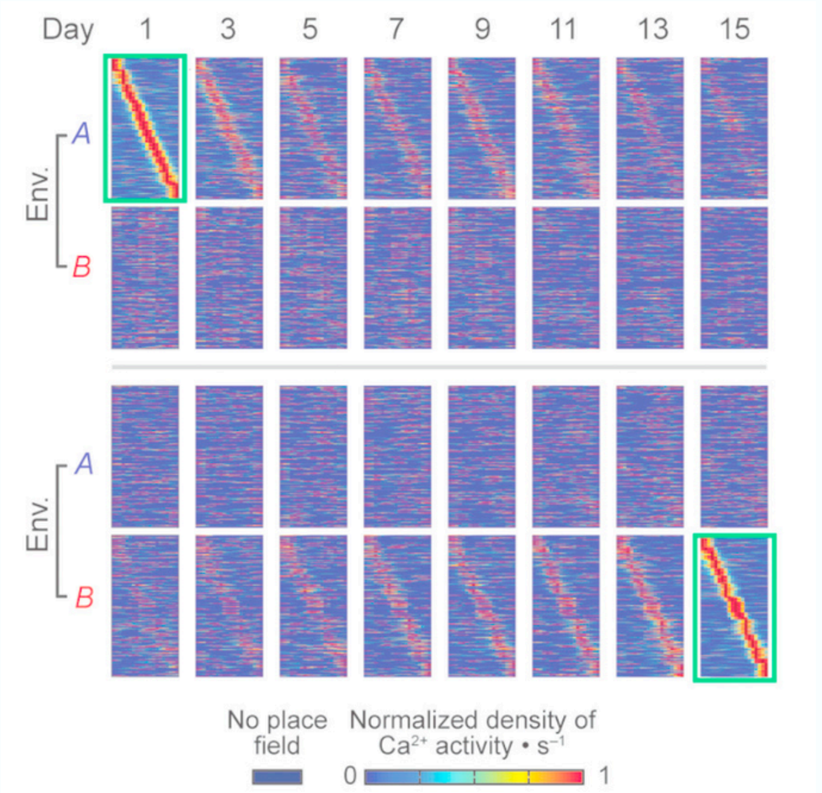


Tracking hippocampal ensemble dynamics across space and time

The hippocampal code is never static



Distinct environments recruit distinct hippocampal representations

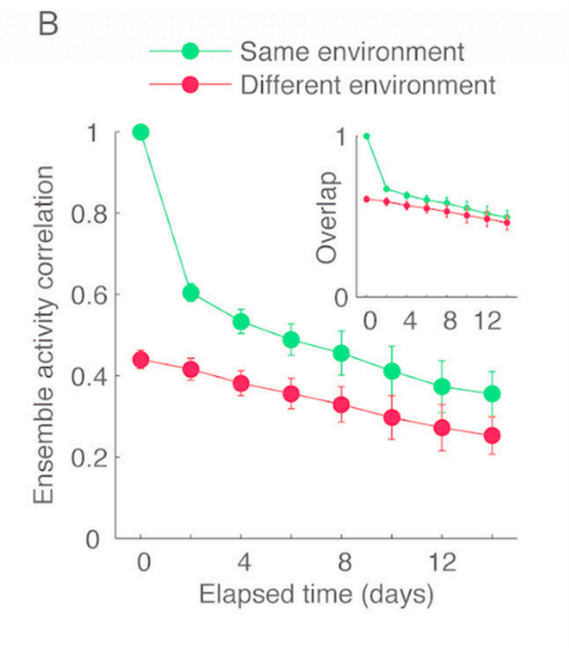


Ensemble activity gradually decorrelates across time within each environment

Hippocampal ensemble representations gradually evolve across days despite stable spatial environments

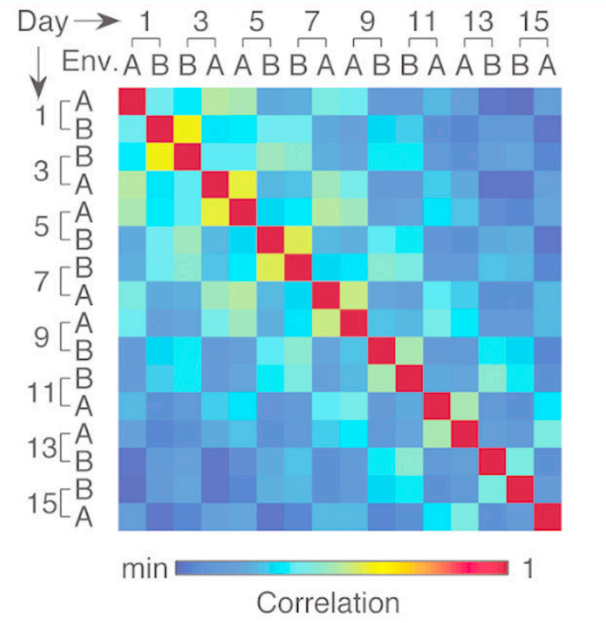
Each day develops a unique neural signature

Within-environment temporal structure



Temporal coding is partially global

Global temporal corrrelation matrix



Hippocampal ensembles contain decodable temporal information
Rubin, A. et al. (2015)

The Hippocampus is inherently dynamic & plastic

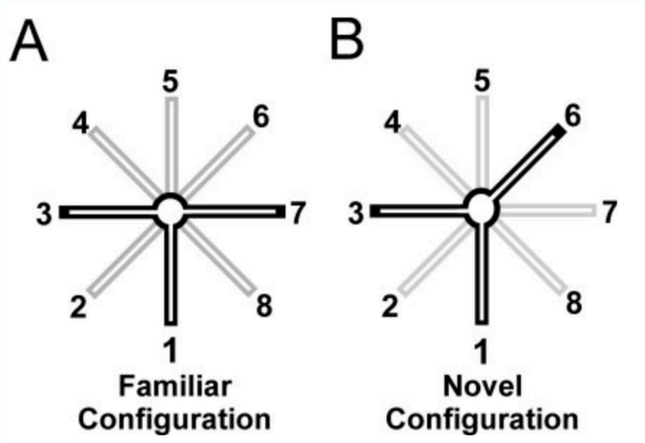
Frank, L. M. et al. (2006)

Classical View

Place fields were thought to represent stable spatial maps

Frank's Question

How stable are hippocampal representations during learning?



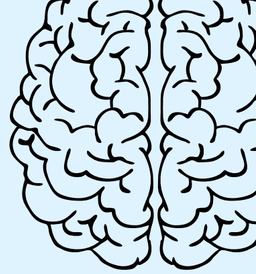
THE HIPPOCAMPUS OPERATES IN A HIGHLY PLASTIC ENCODING STATE.

During novel experience:

1. **Rapid place-field reorganization**
2. **fire-rate changes**
3. **Dynamic remapping**

Plasticity may enable long-term ensemble drift

Stability and flexibility coexist



Spatial Coding
Place Cells
“WHERE”

Temporal Order
Temporal Context
Manns et al.

Time cells
Gap Bridging
MacDonald et al.

Long- Term
Temporal Coding
Neural Timestamaps
Rubin et al.

Plasticity
Continuous
Reorganization
Frank et al.

The hippocampus constructs the timeline of experience

References

Rubin, A. et al. (**2015**)
“Hippocampal ensemble dynamics timestamp events in long-term memory”
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Frank, L. M. et al. (**2006**)
“Hippocampal and cortical place cell plasticity”
Hippocampus

THANKYOU!